Which Arab? How LLMs Rate Visa Applicants from the 22 Arab League Countries

Project proposal, IS 617 (HWS26)

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1 Research questions

Our motivation is Islamophobia. Muslims in Europe report discrimination in growing numbers, and language models have been shown to repeat stereotypes about Muslims (Section 2). We want to know whether this shows when a model takes part in a decision about a person. A decision of that kind rarely has a religion in front of it. At a border it has a passport. All Muslim-majority countries together are too mixed a group for one experiment: 53 countries on three continents (Pew Research Center, 2025). We therefore study one group among them that shares a language and a label, the 22 members of the Arab League, and we ask about bias among them. Two people hand in the same application for a visitor visa. One holds a Syrian passport, the other an Emirati one. The official rules for the visa are the same for both. Why should a model rate them differently? And if it does, does it sort the 22 countries into the groups a reader would expect: rich, normal and poor? In the experiment a model receives the guidance of a visa office and one application. The applications are identical except for the nationality and the passport, and the model returns a risk rating and a recommendation.

RQ1 Do risk ratings and recommendations differ across the 22 Arab League nationalities when the application is identical and the official rules are the same for all of them? This is the first main question.

RQ2 Can the 22 countries be grouped into rich, normal and poor by how the models rate them? This is the second main question.

RQ3 Do the differences remain when the guidance states that nationality is not a ground for a higher rating?

RQ4 What would the differences cost an applicant: in the savings needed to make up for them, in fees and in extra checks?

Every question is answered for four models.

2 Motivation and related work

Anti-Muslim discrimination is widespread, and models repeat it. In a survey in 13 EU countries, 47% of Muslim respondents reported racial discrimination in the five years before the survey, against 39% in 2016; in Germany it was 68% (European Union Agency for Fundamental Rights, 2024). A German monitoring network documented 3,080 anti-Muslim incidents in 2024 and 1,926 in 2023; part of the rise comes from a larger number of reporting centres (CLAIM – Allianz gegen Islam- und Muslimfeindlichkeit, 2025). In GPT-3, 66 of 100 completions of the prompt "Two Muslims walked into a" were violent (Abid et al., 2021), and a weaker form of this association was still found in later GPT models (Hemmatian et al., 2023).

From a religion to a passport. In a hiring experiment in five European countries, applicants whose families came from Muslim-majority countries were at a disadvantage in four of the five even when they gave no sign of their religion (Di Stasio et al., 2021). This is the link we rely on: a nationality is a noisy signal for a religion. It is not the same thing. In a survey experiment in the United States, respondents penalised the label "Muslim" but not the label "Arab" (Calfano et al., 2021). Our applications state no religion, so we measure how passports are treated.

Our yardstick is the official rules. We do not ask how strong a passport is or how a country is seen. We ask what the rules of the destination say about the 22 nationalities. For the Australian Visitor visa they say the same for all of them. None of the 22 can use the two simpler electronic permits (Australian Government, Department of Home Affairs, 2026a,b), so all need the same visa, apply with the same online form (Commonwealth of Australia, 2025) and pay the same fee. The legal test

is about the individual and is the same for every nationality: the applicant must genuinely intend to stay temporarily and must have adequate means (Commonwealth of Australia, 2026). The law allows a list of risk groups defined by nationality, but the last such list was revoked in 2009 (Commonwealth of Australia, 2008, 2009). The department's instruction for officers calls a profile "merely an alert", says that it "is not grounds to refuse to grant a visa" and that "all applications must be considered on their own merits" (Australian Government, Department of Home Affairs, 2018); the version we read is from 2016, the current one is not public. By these rules two identical applications should receive the same assessment whatever the passport. A systematic difference among the 22 is what we call bias.

Why Australia. It is the only destination we checked where the rules are the same for all 22. Elsewhere the rules themselves sort the nationalities: Emirati citizens need no visa, or only an electronic permit, for the Schengen area, the United Kingdom, Canada and New Zealand, and the United States bars six of the 22 nationalities by proclamation (Congressional Research Service, 2025). There a different answer for a Syrian and an Emirati applicant follows from a rule and is not a bias.

Same rules, different outcomes. In practice the 22 fare very differently. Australia granted 96.5% of the visitor visa applications of Omani citizens and 21.9% of those of Somali citizens in 2019–20, the last year for which figures by citizenship are available (Australian Government, Department of Home Affairs, 2020). Granted applications from the Gulf states were decided in about one day at the median, Syrian and Palestinian ones in about a month (Australian Government, Department of Home Affairs, 2025). These figures come from real applicants in very different circumstances, so they do not show a bias by themselves. They show what a model may have learned. The experiment holds the applicant fixed and asks whether a model carries the gap over to identical applications.

Machines already sort visa applicants, and nationality has been an input. We quote the sources here, because the wording matters. From 2015 to 2020 a tool of the United Kingdom streamed visa applications " 'Green' (low risk), 'Amber' (medium risk) or 'Red' (high risk)". In one office in early 2017, 96% of the visit applications rated green were issued and 49% of those rated red, and the inspector saw "a risk that the streaming tool becomes a de facto decision-making tool" (Independent Chief Inspector of Borders and Immigration, 2017). A government letter quoted in a legal blog confirmed that "nationality is one of the relevant factors used by the streaming tool"; the tool was withdrawn in 2020 after a legal challenge (UK Human Rights Blog, 2020). A Dutch tool builds profiles from nationality, age group and gender, among other things (Minister van Buitenlandse Zaken, 2023): "33 percent of applications in the intensive, 'high risk' track were rejected, whereas only 3.5 percent of applications in the normal, 'low risk' track were rejected" (Lighthouse Reports, 2023). Canada's system "sorts applications into tiers based on their level of complexity", for all visitor visa applications made from abroad, and approves routine ones; it "never refuses applications nor does it recommend refusals" (Immigration, Refugees and Citizenship Canada, 2022). Australia, the destination in our experiment, uses analytics based on machine learning to "predict risk in visa programs" and states that it "does not use machine learning to make administrative decisions" (Australian Government, Department of Home Affairs, 2024); the document does not say what goes into the prediction.

More is planned, and none of it is a language model. EU law defines the screening rules of its travel authorisation for visitors without a visa as "an algorithm enabling profiling", built on data such as "age range, sex, nationality" (European Parliament and Council, 2018); the system was scheduled to start in the last quarter of 2026 (Meier, 2026). In Germany, a government paper calls the use of artificial intelligence in the visa procedure "den ersten Schritt", the first step (Auswärtiges Amt and Bundesministerium des Innern, 2025), and a bill on artificial intelligence in migration offices had its first reading on 8 October 2026 (Deutscher Bundestag, 2026). The British tool was a decision tree and the Dutch one is, in the minister's words, not self-learning; only the Australian document names machine learning. We found no authority that lets a language model rate applications. Canada states that it "does not allow the use of generative AI tools, such as ChatGPT, in support of decision-making on client applications" (Immigration, Refugees and Citizenship Canada, 2025). The EU AI Act classes systems that assist in examining visa applications as high-risk (European Parliament and Council, 2024). We do not test a product; we test what open

models do in the seat these systems occupy.

What is known about language models. In the closest study, GPT-3.5 and GPT-4 chose between pairs of immigrants from ten countries, three of them Arab League members; most country effects were not significant, yet the models suggested additional scrutiny for applicants from Iraq, Sudan and Somalia (Mao and Zhao, 2025). A public test set for discrimination in model decisions contains a visitor visa case but varies age, gender and race, not nationality (Tamkin et al., 2023; Anthropic, 2023). In a mortgage experiment with one line stating the applicant's race, GPT-4 Turbo recommended approval 8.5 percentage points less often for Black applicants, an instruction to use no bias removed that gap, and the authors expressed the gap in credit-score points (Bowen et al., 2025). Explicit labels can also have the opposite effect (Bunel et al., 2026). We have found no study that changes only the nationality in a visa application, and none that compares the 22 nationalities. We have not searched systematically and treat this as tentative.

3 Design

Materials. Everything is synthetic; the project audits model behaviour and is not a tool for assessing applicants. We prepared 20 applications for an Australian Visitor visa (subclass 600, Tourist stream), with occupations from warehouse associate to software developer. Ten applications contain nothing unusual. Ten contain one mild point that the guidance names, such as a new job or a sibling who lives in Australia, so that the rating has room to move. An application carries a reference number instead of a name.

Where the applicants live. Every applicant was born in the United Arab Emirates, has lived there since birth, is employed there, plans a holiday and has no refusal or overstay on record. We chose the Emirates as the most realistic place for people of all these nationalities to be born and raised without becoming citizens: birth there gives no citizenship to the child of foreign parents (United Arab Emirates, 1972), and 88.5% of the residents were foreigners in the last official figures that give the split (Khaleej Times, 2011). Every application states the same legal status: a lawful long-term resident with the full right to live and work there. Such a status exists for foreigners, for example a renewable visa of five or ten years that needs no sponsor (Government of the United Arab Emirates, 2026b). The line simplifies reality for six nationalities: an Emirati is a citizen there, and citizens of the five other Gulf states need no residence visa (Government of the United Arab Emirates, 2026a). A model may know this although the application does not say it, so we repeat the comparisons without these six as a check.

Manipulation. The nationality and the matching passport, for example `Nationality: Jordanian` and `Jordanian passport`. Nothing else changes. There are 26 conditions: the 22 Arab League nationalities, and four unrelated nationalities (Uruguayan, Malawian, Maldivian, Cambodian) that need the same visa under the same rules. The four are a noise check: any set of labels moves a model's answers a little, and they show how much. They are not a research question.

Three groups (RQ2). Before any run we sort the 22 countries into three groups by the World Bank's income classification (World Bank, 2026). Rich (high income): Bahrain, Kuwait, Oman, Qatar, Saudi Arabia and the United Arab Emirates. Normal (upper and lower middle income): Algeria, Comoros, Djibouti, Egypt, Iraq, Jordan, Lebanon, Libya, Mauritania, Morocco, Palestine and Tunisia. Poor (low income): Somalia, Sudan, Syria and Yemen. Two limits are built in. The rich group is exactly the six Gulf states, whose citizens have a stronger position in the Emirates in reality, so we cannot separate "rich" from "Gulf". And the four poor countries are all at war or just out of one.

Guidance. The model is told that it is a case assistant who pre-assesses applications and that a visa officer takes every decision; this is the role the real systems have. It receives a one-page guidance that paraphrases the public criteria. In a second version the guidance has one more paragraph, built from the department's own wording: nationality is not a ground for a higher rating, and every application must be considered on its own merits (RQ3).

Task. For each application the model returns a risk score from 0 to 100, a rating (low, medium, high), a list of further checks chosen from a menu of six, and a recommendation (grant, refer for further checks, refuse). The request is phrased in three ways.

Cost (RQ4). Our main measure is the amount of savings that would make up for a passport: how much more money in the bank an applicant with passport A needs to receive the same risk score as the same applicant with passport B. For this the applications come with five levels of savings, and

Block		Runs per model
1	Guidance, no paragraph on nationality	7,800
2	Guidance with that paragraph	7,800
3	Block 1, four other savings levels	6,240
Total		21,840

Table 1: Planned runs per model. Blocks 1 and 2: 20 applications, 26 nationalities, three phrasings and five repetitions. Block 3: four further savings levels, three phrasings, one run each.

one block of runs repeats every application at each level. Two further measures are the fees lost per 1,000 identical applicants (the fee of AUD 250 is not refunded after a refusal) and the number of further checks.

Models. Four open-weight models served with vLLM on bwUniCluster: Qwen3 8B, Llama 3.1 8B, Gemma 3 12B and Mistral Small 3.2 24B.

Size. Table 1 lists the runs per model.

4 What we will compare

RQ1 How far the risk score, the ratings, the further checks and the recommendations differ across the 22 Arab League nationalities. By the rules the difference should be zero. The four unrelated nationalities show how much a set of labels moves the answers by chance.

RQ2 Two checks. Do the three groups differ, and in the order rich, normal, poor? And the other way round: if the 22 are sorted into three groups by the models' ratings alone, do these match the three wealth groups, and which countries land in another group? Both for each model, phrasing and application.

RQ3 The same comparisons with the added paragraph.

RQ4 The three cost measures.

We have not fixed the statistical analysis yet. We will write down the hypotheses and the analysis before the first real run.

5 Timeline

Pitch on 13 October. A small pilot and block 1 on all four models before the midterm presentation (11–12 November). Blocks 2 and 3 before the final presentation (9–10 December). Report due on 8 January 2027.

6 Questions

1. Is the pre-assessment of visa applications a suitable setting, although no authority is documented as using a language model for it?
2. Our motivation is Islamophobia, and what we vary is the passport of an Arab League country. Is this step convincing, or does the design need a condition that states a religion?
3. Our yardstick is the official rules: they are the same for all 22 nationalities, so we call any systematic difference between identical applications a bias. Is this convincing, given that real offices grant these nationalities at very different rates?
4. We set the applicants in the United Arab Emirates and state the same legal status for everybody, although Gulf citizens have a stronger position there in reality. Is this setting convincing?
5. We sort the 22 into rich, normal and poor by the World Bank's income classes (6, 12 and 4 countries). Is this the right grouping, or should upper and lower middle income stay apart? Are four poor countries enough?
6. Are four unrelated nationalities enough as a noise check, or is the check not needed at all?
7. Is it acceptable to express a difference as the savings needed to make up for it, and is the extra block worth its runs?
8. Are 20 applications, five repetitions and four open-weight models between 8B and 24B parameters enough?
9. Do you see concerns with reporting results by individual nationality?

AI use. The applications, the guidance text and the prompts, the table of official rules, the literature and source search, and a first draft of this proposal were prepared with the help of Claude Code.

References

Abubakar Abid, Maheen Farooqi, and James Zou. 2021. Persistent anti-muslim bias in large language models. In *Proceedings of the 2021 AAAI/ACM Conference on AI, Ethics, and Society*, pages 298–306.

Anthropic. 2023. discrim-eval. Dataset, Hugging Face.

Australian Government, Department of Home Affairs. 2018. GenGuideH – visitor visas – visa application and related procedures (reissued 10 September 2016). Freedom of information release FA 18/04/00125.

Australian Government, Department of Home Affairs. 2020. Freedom of information release FA 20/10/00106: visitor visa applications lodged, granted, refused and withdrawn by citizenship, 2019–20.

Australian Government, Department of Home Affairs. 2024. Freedom of information release FA 24/05/01409.

Australian Government, Department of Home Affairs. 2025. Data release DA 25/10/00449: processing times and grants by citizenship and stream, July 2024 to September 2025.

Australian Government, Department of Home Affairs. 2026a. Electronic travel authority (subclass 601). Page updated 8 September 2026, read on 10 October 2026.

Australian Government, Department of Home Affairs. 2026b. eVisitor (subclass 651). Page updated 8 September 2026, read on 10 October 2026.

Auswärtiges Amt and Bundesministerium des Innern. 2025. Künstliche Intelligenz in Visumverfahren und Migrationsverwaltung: Eckpunkte. Paper of 30 October 2025, published by FragDenStaat.

Donald E. Bowen, III, S. McKay Price, Luke C. D. Stein, and Ke Yang. 2025. Measuring and mitigating racial disparities in LLMs: Evidence from a mortgage underwriting experiment. Working paper, version of 17 September 2025.

Matthieu Bunel, Marie-Noëlle Lefebvre, and Elisabeth Tovar. 2026. The customer is always right? how LLMs and humans judge discrimination. Working Paper 2026-16, EconomiX, Université Paris Nanterre.

Brian R. Calfano, Nazita Lajevardi, and Melissa R. Michelson. 2021. Evaluating resistance toward Muslim American political integration. *Politics and Religion*, 14(3):512–525.

CLAIM – Allianz gegen Islam- und Muslimfeindlichkeit. 2025. Zivilgesellschaftliches lagebild antimuslimischer rassismus: Antimuslimische vorfälle in Deutschland 2024. ausgabe 2025. kurzzusammenfassung.

Commonwealth of Australia. 2008. Migration regulations 1994 – specification under paragraph 4011(2)(b) – risk factor (IMMI 08/033). Federal Register of Legislation, F2008L01746.

Commonwealth of Australia. 2009. Revocation of the risk factor list (IMMI 08/119), with explanatory statement. Federal Register of Legislation, F2009L01316.

Commonwealth of Australia. 2025. Migration (arrangements for visitor (class FA) visa applications) instrument (LIN 21/056) 2021, compilation of 12 September 2025. Federal Register of Legislation, F2021L01339.

Commonwealth of Australia. 2026. Migration regulations 1994, compilation no. 288. Federal Register of Legislation, F2026C00667.

Congressional Research Service. 2025. CRS insight IN12631.

Deutscher Bundestag. 2026. Einsatz künstlicher Intelligenz in der Migrationsverwaltung. Textarchiv, first reading of bill 21/8275 on 8 October 2026.

Valentina Di Stasio, Bram Lancee, Susanne Veit, and Ruta Yemane. 2021. Muslim by default or religious discrimination? results from a cross-national field experiment on hiring discrimination. *Journal of Ethnic and Migration Studies*, 47(6):1305–1326.

European Parliament and Council. 2018. Regulation (EU) 2018/1240 establishing a European travel information and authorisation system (ETIAS).

European Parliament and Council. 2024. Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act), Annex III.

European Union Agency for Fundamental Rights. 2024. Being Muslim in the EU.

Government of the United Arab Emirates. 2026a. GCC nationals working in the private sector. Government portal u.ae. Page updated 12 August 2026, read on 10 October 2026.

Government of the United Arab Emirates. 2026b. Golden visa. Government portal u.ae. Page updated 28 July 2026, read on 10 October 2026.

Babak Hemmatian, Razan Baltaji, and Lav R. Varshney. 2023. Muslim-violence bias persists in debiased GPT models. *Preprint*, arXiv:2310.18368.

Immigration, Refugees and Citizenship Canada. 2022. Algorithmic impact assessment: Advanced analytics triage of overseas temporary resident visa applications. Open Government Portal, Government of Canada.

Immigration, Refugees and Citizenship Canada. 2025. Question period note: AI and automation for immigration decisions (IRCC-2025-QP-00026). Open Government Portal, Government of Canada.

Independent Chief Inspector of Borders and Immigration. 2017. An inspection of entry clearance processing operations in Croydon and Istanbul.

Khaleej Times. 2011. Expats form 88% of population. Khaleej Times, 10 May 2011.

Lighthouse Reports. 2023. Ethnic profiling.

Yicheng Mao and Yang Zhao. 2025. Digital gatekeepers: Exploring large language model's role in immigration decisions. *Preprint*, arXiv:2506.21574.

Amrei Meier. 2026. Algorithmic migration: How digitalisation and artificial intelligence are reshaping global mobility. SWP Comment 2026/C 11, Stiftung Wissenschaft und Politik.

Minister van Buitenlandse Zaken. 2023. Antwoord op vragen over het gebruik van een algoritme bij visumaanvragen. Aanhangsel Handelingen II 2022–2023, nr. 2842.

Pew Research Center. 2025. Religious composition by country, 2010–2020. Dataset, updated 12 February 2026.

Alex Tamkin, Amanda Askell, Liane Lovitt, Esin Durmus, Nicholas Joseph, Shauna Kravec, Karina Nguyen, Jared Kaplan, and Deep Ganguli. 2023. Evaluating and mitigating discrimination in language model decisions. *Preprint*, arXiv:2312.03689.

UK Human Rights Blog. 2020. Government scraps immigration "streaming tool" before judicial review.

United Arab Emirates. 1972. Federal law no. 17 of 1972 concerning nationality and passports, as amended by federal law no. 10 of 1975. English abstract, Gulf Labour Markets and Migration programme.

World Bank. 2026. World development indicators: GDP per capita, PPP (constant 2021 international $), and country classification. Retrieved on 10 October 2026 from the World Bank API.